Fiscal Year:	FY 2015	**Task Last Updated:**	FY 05/05/2015
PI Name:	Durian, Douglas J. Ph.D.		
Project Title:	FOAM--Foam Optics and Mechanics (The Melting of Aqueous Foams)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-6396	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	1995 Microgravity Fluid Physics NRA-94-OLMSA-05: Research and Flight Experiment Opportunities
Start Date:	07/21/2014	**End Date:**	07/20/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX14AM99G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	ED. NOTE (May 2015): Continuation of "The Melting of Aqueous Foams"; previous grant numbers NNX07AP20G ; NNX07AK60G ; NNC04GB61G. The unusual elastic character of foams will be quantified macroscopically by measurement of the shear stress as a function of static shear strain, shear strain rate, and time following a step strain; such data will be analyzed in terms of a yield stress, a static shear modulus, and dynamical time scales. Microscopic information about bubble packing and rearrangement dynamics, from which these macroscopic non-Newtonian properties presumably arise, will be obtained non-invasively by novel multiple-light scattering diagnostics such as diffusing-wave spectroscopy (DWS) and speckle-visibility spectroscopy (SVS). Bubble size and homogeneity will be studied by diffusely transmitted light (DTS). Liquid fraction will be measured by electrical conductivity. Quantitative trends with materials parameters, such as average bubble size, and liquid content, will be sought in order to elucidate the fundamental connection between the microscopic structure and dynamics and the macroscopic rheology. There are three all-encompassing objectives motivating the FOAM project. The first concerns multiphase flows and represents the “mechanics” aspect alluded to in the F.O.A.M. acronym: To understand how foams and bubbly liquids deform and flow in response to applied forces. The second objective concerns multiphase separations: To understand how foams and bubbly liquids evolve with time via a combination of liquid flow between bubbles, of gas diffusion across the film separating neighboring bubbles, and of direct coalescence via film rupture. For both flows and separations, some level of understanding has been established for very dry foams, and also for liquids with relatively few bubbles; however, we are more ignorant of the crucial intermediate regime of very wet foams and very bubbly liquids. This regime defies Earth-bound experiments. The third, overall, objective is to transfer this new understanding of multiphase flows and separations into ASL technology for NASA's strategic exploration mission. The utility and fascination of foams are derived largely from the surprising fact that they have a solid-like elastic character in spite of being mostly gas with a few percent volume fraction of liquid, but can nevertheless flow under shear. The physical origin of such unusual rheology in terms of microscopic structure and dynamics is poorly understood and remains a subject of basic scientific interest to physicists, chemists, and chemical engineers. The proposed research promises important new insight into these issues, and could also have significant consequences for our understanding of flow in other dense randomly-packed systems such as emulsions, colloidal suspensions, slurries, bubbly liquids, and granular materials. Furthermore, all foam applications are empirically based and the proposed research may generate valuable fundamental guidance for the development of materials with more desirable rheology and better stability. The value of the FOAM project in terms of NASA’s exploration mission lies with Advanced Life Support. Water recovery and recycling are absolutely central to ASL, and require fine control of multiphase flows and separations. The very heart of the FOAM project is to bring new technology and understanding to these problems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Foams are important for a clean environment. Firstly, their presence in natural waters signals the presence of surface-active impurities and a need for clean-up. Secondly, they are useful themselves in the cleanup process since many impurities can be removed by injecting gas and skimming off the resulting foam. Relatedly, study of foams brings specific insight into multiphase flow and separation phenomena of direct importance for water recovery and recycling operations.
Task Progress:	During this cycle my FOAM group consisted of one graduate student, Mr. Anthony Chieco. The focus of Anthony’s PhD research is on ground-based studies of the structure and time evolution of coarsening foams, in support of the planned flight experiments with FSL experiment container SOFT MATTER DYNAMICS (formerly FOAM-C). This is being developed by the European Space Agency. It is currently in phase C/D, with a target launch date in the second half of 2017. A CDR (critical design review) has not yet been scheduled. The FOAM focus is on structure, stability, and temporal evolution, for sequences of progressively wetter foams. These cannot be studied on Earth due to gravitation drainage, which increases with wetness and effectively prevents study of behaving when the bubbles become nearly spherical and unjammed. At Penn, we have made progress in several areas during the first year of renewed funding (cut off April 2013—July 2014). [1] High-precision measurements of bubble shape parameters for wet foams during evolution. [2] Monte-Carlo foam reconstruction technique. [3] Introduction of a hyperuniformity disorder length for characterizing the large-scale foam structure. These methods will all be brought to bear on anticipated flight experiment data. In addition, two papers appeared in 2014: one on multiple-light scattering probes of foam structure and dynamics – also to be used for the flight experiment – and another on a novel method of measuring foam rheology (a topic proposed but ultimately abandoned for the current fight experiments). These topics are described in detail in the technical report, along with current plans for the on-going topics.
Bibliography Type:	Description: (Last Updated: 06/12/2025)
Abstracts for Journals and Proceedings	Chieco A, Roth AE, Dreyfus R, Torquato S, Durian DJ. "Hyperuniformity length in experimental foam and simulated point patterns." Presented at March Meeting of the American Physical Society, San Antonio, TX, March 2-6, 2015. Bulletin of the American Physical Society. 2015 Mar;60(1):J50.04. http://meetings.aps.org/link/BAPS.2015.MAR.J50.4 ; accessed 5/5/15. , Mar-2015
Articles in Other Journals or Periodicals	Jones CD, Nordstrom KN, Durian DJ. "Rheology of nearly ideal 3d foams." arXiv:1404.2857. Submitted April 2014. http://arxiv.org/abs/1404.2857 , Apr-2014
Articles in Peer-reviewed Journals	Hohler R, Cohen-Addad S, Durian DJ. "Multiple light scattering as a probe of foams and emulsions." Current Opinion in Colloid and Surface Science. 2014 Jun;19(3):242-52. http://dx.doi.org/10.1016/j.cocis.2014.04.005 , Jun-2014